

AISHWARYA SINGH

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LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineer II with 3+ years of experience designing and delivering scalable backend services, APIs, and real-time workflow platforms for equity-linked structured products at Bank of America. Skilled in Python, TypeScript, microservices, event-driven systems, React, automated testing, CI/CD, and Azure, with hands-on experience applying machine learning to derivatives pricing.

EXPERIENCE

Bank of America | Singapore

Software Engineer II, Equity Linked Technology - Structured Products

Jul 2025 - Present

- Architected and delivered a unified *Quote API* that persists pricing results and quote metadata across desks, centralizing business rules and reducing quote-capture time by **79%**.
- Scaled backend and API workflows to support **120-150** completed trade bookings per week, up from **2-10**, through service and integration improvements.
- Cut end-to-end quote-pricing compute time by **40%** across supported workflows by optimizing pricing-engine processing and downstream integration paths.
- Owned delivery and APAC production support for pricing and booking services used across APAC, EMEA, and the Americas, triaging incidents and user-reported issues for sales, structurers, and traders.

Graduate Technology Analyst, Equity Linked Technology

Jul 2023 - Jul 2025

- Expanded a React/TypeScript real-time quote dashboard and blotter from **150K** to **1M** stored quotes by scaling AMPS event processing and live frontend-update workflows.
- Re-engineered Python microservice APIs powering quote retrieval, improving frontend query performance by **60%** and accelerating trader and sales workflows.
- Strengthened the two-week production release cycle through automated testing, CI/CD quality gates, coding standards, and peer review across pricing, booking, and client-connectivity workflows.

EDUCATION

Nanyang Technological University | Singapore

2019 - 2023

Bachelor of Engineering (Honours), Computer Engineering

Academic focus: Data Science & Artificial Intelligence

PROJECTS & RESEARCH

ML-Pricer - Machine Learning Pricing Framework | Personal Project

2024 - Present

GitHub | Python, FastAPI, Streamlit, LightGBM, Monte Carlo Simulation

- Engineered an end-to-end framework that trains surrogate models to approximate Monte Carlo valuations for Phoenix autocallables, accumulators, and barrier products.
- Achieved **R-squared above 0.98** on tested configurations and **20-100x faster inference** than Monte Carlo, with benchmarks spanning predictive accuracy, latency, and runtime.
- Exposed pricing, model-status, and scenario-analysis workflows through FastAPI and Streamlit, supported by model caching and automated tests.

Cultivating Navigational Autonomy in the Visually Impaired: A Novel Approach with VirtualEYE

2024

IEEE Conference on Artificial Intelligence (CAI) | First Author

- Engineered and evaluated VirtualEYE, an Android indoor-navigation system combining BLE-beacon localization, motion sensors, computer vision, and multimodal feedback; reduced navigation time by **20.48%** and **37.34%** across visual and non-visual trials, respectively.

TECHNICAL SKILLS

Engineering: Python, Java, TypeScript, SQL, REST APIs, Microservices, API Design, Event-Driven Systems, AMPS Messaging, FastAPI, React

Operations & AI/ML: CI/CD, Automated Testing, Code Review, Production Support, Incident Triage, Docker, Microsoft Azure, LightGBM, TensorFlow Lite, Large Language Models (LLMs)

Markets: Structured Products, Derivatives Pricing, Quote Lifecycle, Booking Systems